

TCP/IP Suite

1. Create a side-by-side comparison table showing at least 4 key differences between the OSI Model and the TCP/IP Model, including the number of layers, layer names, and where protocols like HTTP, TCP, and IP fit.

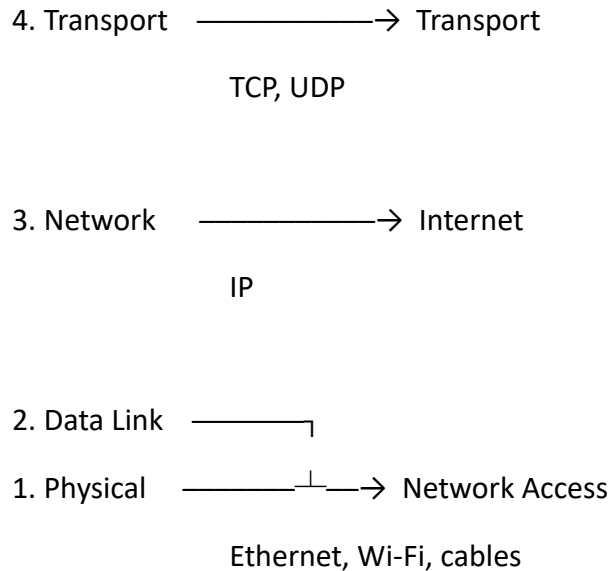
->OSI Model vs TCP/IP Model

OSI Model	TCP/IP Model
1. Number of layers: 7 layers	Number of layers: 4 layers
2. Layer names: Application, Presentation, Session, Transport, Network, Data Link, Physical	Layer names: Application, Transport, Internet, Network Access
3. Application protocols: HTTP/HTTPS are at Application Layer (7)	HTTP/HTTPS are at Application Layer
4. Transport protocols: TCP and UDP are at Transport Layer (4)	TCP and UDP are at Transport Layer
5. Network protocol: IP is at Network Layer (3)	IP is at Internet Layer
6. Data Link & Physical: Separate layers for Ethernet, cables, signals, etc.	Combined into the Network Access/Link Layer
7. Main purpose: Mainly a conceptual/reference model used to understand networking	A practical protocol suite/model used by the Internet

Layer Mapping

OSI MODEL (7 Layers) TCP/IP MODEL (4 Layers)

7. Application ————
 6. Presentation ————
 5. Session ————+——> Application
 | HTTP, HTTPS, DNS



Key point: The OSI model divides networking into **7 distinct layers**, while TCP/IP combines some of those functions into **4 broader layers**. Both models help explain how protocols such as **HTTP, TCP, and IP** work together to transmit data.

2. Open Wireshark, start a capture on your active network interface, and visit any website (like www.youtube.com) in your browser. Stop the capture and find a TCP packet in the list. Write down the source IP, destination IP, source port, destination port, and the TCP flag(s) set for that packet.

->I can't see or capture packets from your computer directly, so use the following steps to record the **actual values from your Wireshark capture**.

Wireshark TCP Packet Task

1. Open **Wireshark**.
2. Select your active interface, such as **Wi-Fi** or **Ethernet**.
3. Click **Start Capturing**.
4. Open your browser and visit youtube.com.
5. After the page loads, return to Wireshark and stop the capture.
6. In the display-filter box, enter:

tcp

7. Select any TCP packet from the packet list.

8. Expand **Transmission Control Protocol** in the Packet Details pane.

Record These Values

Information	Your Wireshark Result
Source IP	_____
Destination IP	_____
Source Port	_____
Destination Port	_____
TCP Flags	_____
Website	YouTube
Protocol	TCP

For example, a TCP packet might show a destination port of **443** for HTTPS traffic, with flags such as **ACK**, **SYN**, or **PSH, ACK**. **Do not copy these as your result**—record the values actually displayed in your selected packet.

 **Documentation:** Take a screenshot showing the selected TCP packet and the expanded **IPv4** + **TCP** sections.

3. In Wireshark, filter packets by 'udp' and find a DNS query. Identify and note which TCP/IP layer each of the following fields belongs to: source IP, destination IP, source port, destination port, DNS query name

->Wireshark UDP/DNS Analysis

1. Open your Wireshark capture.
2. Apply the filter:

udp

3. Find a packet containing **DNS**. You can also use:

dns

4. Select a DNS query and expand the **Internet Protocol**, **User Datagram Protocol**, and **Domain Name System** sections.

Field-to-TCP/IP Layer Mapping

Field	Example	TCP/IP Layer
Source IP	192.168.1.10	Internet Layer
Destination IP	8.8.8.8	Internet Layer
Source Port	54321	Transport Layer
Destination Port	53	Transport Layer

DNS Query Name example.com **Application Layer**

Explanation

- **Source IP & Destination IP → Internet Layer:** IP addresses identify the sending and receiving hosts.
- **Source & Destination Ports → Transport Layer:** UDP uses port numbers to identify the application/service communicating.
- **DNS Query Name → Application Layer:** DNS is an application-layer protocol, and the query name identifies the domain being requested.

Documentation: Take a screenshot of the selected DNS packet with the **Internet Protocol**, **UDP**, and **DNS** sections expanded.

4. Analyze any captured IPv6 packet in Wireshark and list the differences you observe in the IP header compared to an IPv4 packet.

Hint: Look for differences in address format, header length, and any extra fields present in IPv6.

->IPv4 vs IPv6 Header Analysis in Wireshark

Open your Wireshark capture and select an **IPv6** packet. Expand **Internet Protocol Version 6** and compare it with an IPv4 packet's **Internet Protocol Version 4** section.

Feature	IPv4	IPv6
Address format	32-bit, e.g. 192.168.1.10	128-bit hexadecimal, e.g. 2001:db8::1
Header size	Variable: 20–60 bytes	Fixed: 40 bytes
Address fields	Source + Destination IPv4 addresses	Source + Destination IPv6 addresses
Header Length field	Present (IHL)	Not present
TTL / Hop Limit	TTL	Hop Limit
Protocol / Next Header	Protocol field	Next Header field
Fragmentation	Fragmentation information can be in the main IPv4 header	Fragmentation uses an extension header
Checksum	IPv4 header checksum is present	No IPv6 header checksum
Extra/extension information	Options are included in the IPv4 header	Optional information is handled through extension headers

Main Differences Observed

1. **IPv6 addresses are much longer** because they use 128 bits instead of IPv4's 32 bits.
2. **IPv6 has a fixed 40-byte base header**, while IPv4 has a variable-length header.
3. IPv6 removes the **IHL** and **header checksum** fields found in IPv4.
4. IPv6 uses **Hop Limit** instead of IPv4's TTL.
5. IPv6 uses **extension headers** for optional information such as fragmentation.

 **Documentation:** Take a screenshot showing the expanded **Internet Protocol Version 6** section and, if possible, another screenshot of an IPv4 packet for direct comparison.